

AI TOOL PROMPT ENGINEERING REPORT

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Computer Science – 2B
21st Century IT Skills – 16-607



AI Tool Overview

Tool Used: Windsurf (Formerly Codeium)

Purpose: AI-First Code Editor

Motive of usage: I often use Windsurf but never considered prompting my prompt. By the time I prompt in Windsurf, I'm already looking for something specific be it a bug or something to improve, I hardly let it control the overall direction of the codebase. I wanted to experience what it would be like to let AI handle frontend design for once, having it done in two layers (prompting the prompt) is a bonus.

First prompt: "Keeping the overall flow of the code, redesign it in your own way to modern standards. Keep only the fonts; the shapes, the background, the icons, go wild with it."



Second (AI-Enhanced) Prompt: “””Keep the overall structure and flow of the existing HTML/Tailwind code intact, including the pixel font and the four-icon layout. However, redesign the page to meet current modern web standards (accessibility, responsiveness, performance, micro-interactions).

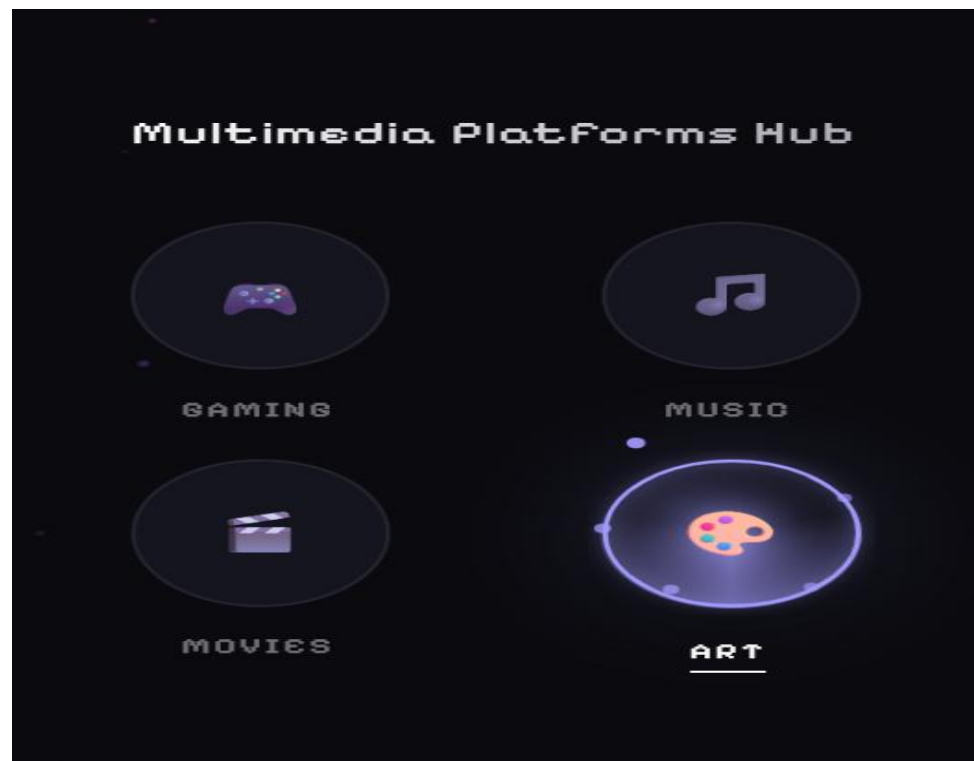
Preserve:

- Pixel font style
- Four circular icons (one per platform)
- Dark background

Redesign with creativity:

- Shapes, background treatment, icon styles, hover/focus states, and layout rhythm
- Add subtle, smooth animations (e.g., icon lift, glow, scale, or background particle effect)
- Include a functional or conceptual layer: each circle should feel like a distinct ‘media hub’ (e.g., animated radial progress, sound wave pulse, or dynamic gradient on hover)
- Improve semantic HTML and ARIA labels for screen readers
- Add a responsive grid or flex layout that stacks gracefully on mobile
- Optional: small interactive label or tooltip that appears on hover/tap showing the platform name
- Maintain dark mode natively (no flash of light background)
- Ensure smooth hardware-accelerated transitions

Go wild within those constraints—modern, bold, immersive, but still usable.”””



Analysis

Given my non-prompted output, I was satisfied with the first prompt already, it kept my aesthetic choice and didn't overcomplicate beyond my expectations; it still uses the Tailwind framework and emphasized my coloring choices more. (Code References in GitHub Repo).

It's difficult to see from the image preview pasted above but the enhanced prompt is very advanced, much more than I can think of if it were me. I do see issues with it however, such as conciseness, readability, and structure being completely incomprehensible. It completely ignored the Tailwind framework I did and even utilized JavaScript just to have more advanced animations. If I had to choose, I'd pick my initial prompt as in jargon, this kind of code would be considered a "Write-only code."

Preference aside, I've noticed that due to the longer prompt in the enhanced version, the model had to split up its tasks (as seen in the Appendix below). Yet even then, because there were too many features clamped in a single request, the model has probably hallucinated several times and over-corrected itself, satisficing with "what works" with as I observed- abandoning Tailwind and opting to use JavaScript. In contrast, though my initial prompt was very basic and broad, the model was able to follow and add its own inputs because it had no chance to over-correct itself and hallucinate between two tasks. Do note that these prompts were done across different context windows and weren't personalized.

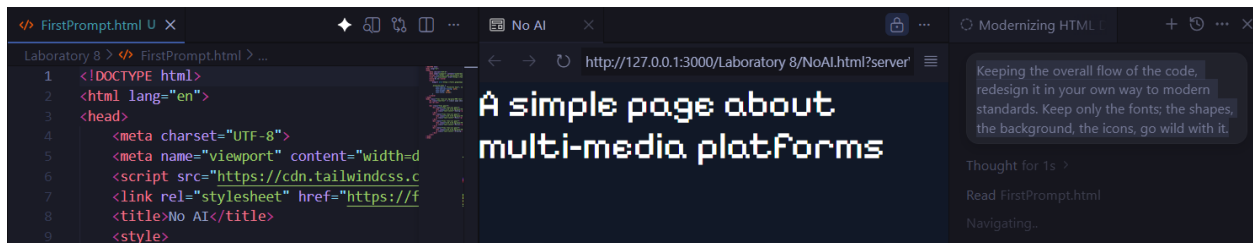
Ergo, from what I observe, concise prompts do help but the "detailed thinking" of a certain model will always be shaped by its core pattern from its dataset, as observed as both prompt outputs were similar in form but chased "uniqueness" through "pretending" to be more complex. There is also another observation I have where despite being concise, longer prompts doesn't necessarily mean an improved outcome, in fact, the larger it is, the more prone the model will hallucinate.

Appendix

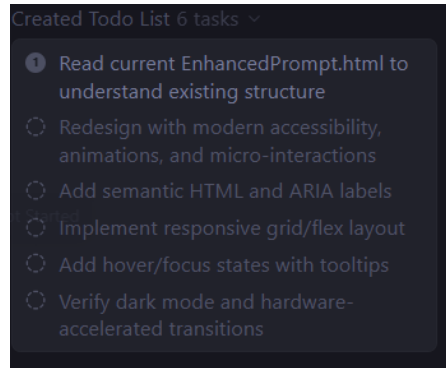
Initial Code (No Prompts):



First Prompt



Created a TODO List due to the volume of requests in the AI-Assisted Prompt



Prompt on DeepSeek: “””I have a raw HTML file with a Tailwind script. What I've done is simple, a dark background, a header with a pixel font and four circles housing an icon each. The theme is "Multi-media platforms."

My prompt to improve the code was “Keeping the overall flow of the code, redesign it in your own way to modern standards. Keep only the fonts; the shapes, the background, the icons, go wild with it.”

Improve my prompt to be more specific, modernized, and have more ideas than just visual reframing.”””